

Fully typed neural networks with lean

Who uses PyTorch here?

Looks familiar?

```
File "/home/danie/codice/lean-cnn/pytorch/.venv/lib/python3.12/site-pack
", line 1778, in _wrapped_call_impl
    return self._call_impl(*args, **kwargs)
    ~~~~~^~~~~~
File "/home/danie/codice/lean-cnn/pytorch/.venv/lib/python3.12/site-pack
", line 1789, in _call_impl
    return forward_call(*args, **kwargs)
    ~~~~~^~~~~~
File "/home/danie/codice/lean-cnn/pytorch/mlp.py", line 15, in forward
    x = self.layer2(x)
    ~~~~~^~~~~~
File "/home/danie/codice/lean-cnn/pytorch/.venv/lib/python3.12/site-pack
", line 1778, in _wrapped_call_impl
    return self._call_impl(*args, **kwargs)
    ~~~~~^~~~~~
File "/home/danie/codice/lean-cnn/pytorch/.venv/lib/python3.12/site-pack
", line 1789, in _call_impl
    return forward_call(*args, **kwargs)
    ~~~~~^~~~~~
File "/home/danie/codice/lean-cnn/pytorch/.venv/lib/python3.12/site-pack
", line 134, in forward
    return F.linear(input, self.weight, self.bias)
    ~~~~~^~~~~~
RuntimeError: mat1 and mat2 shapes cannot be multiplied (32x30 and 29x1)
```

no more with fully typed neural networks



Formalizing Neural Networks in Lean

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TorchLean formalizes neural network infrastructure in Lean 4, connecting typed tensor and layer specifications, runnable training examples, graph IR semantics, floating-point contracts, CUDA trust boundaries, and artifacts that Lean checkers can inspect.

The goal is a practical bridge between modern ML workflows and formal reasoning: models can be executed, lowered, inspected, imported from PyTorch-style pipelines, and checked against explicit mathematical contracts.

<https://lean-dojo.github.io/TorchLean/>



What is Lean?

- Functional programming language and interactive theorem prover
- Every expression has a precise type — the compiler is also a proof checker
- Dependent types let you encode invariants directly in the type system
- Developed at AWS and Microsoft Research; used in formal mathematics (Mathlib)
- Growing ecosystem for verified, type-safe scientific computing

<https://lean-lang.org/>

- A Lean library providing access to PyTorch functionality
- Lives inside Lean's strong type system — with strong static guarantees
- No way to code a network with incompatible layers: shape errors caught at compile time
- Many more guarantees on the software: memory safety, totality, and formal correctness

```
-- MLP with 1 hidden layer of 30 neurons that uses ReLU activation function
def mkModel : nn.M (nn.Sequential 10 1) :=
  nn.sequential![
    nn.linear 10 30 Spec.Shape.scalar,
    nn.relu,
    nn.linear 30 1 Spec.Shape.scalar
  ]
```

Lean gets it

```
-- Input shape: one 10-D feature vector per sample.
abbrev  $\sigma$  : Spec.Shape := NN.Tensor.Shape.Vec 10

-- Output shape: length-1 vector (TorchLean never uses bare scalars)
abbrev  $\tau$  : Spec.Shape := NN.Tensor.Shape.Vec outDim

-- MLP with 1 hidden layer of 30 neurons that uses ReLU activation
def mkModel : nn.M (nn.Sequential  $\sigma$   $\tau$ ) :=
  nn.sequential! [
    -- nn.linear inDim 30      Spec.Shape.scalar,
    nn.linear inDim 29      Spec.Shape.scalar,
    nn.relu,
    nn.linear 30      outDim Spec.Shape.scalar
  ]
```

▼ All Messages (⊗ 1)

▼ Model.lean:36:2

Application type mismatch: The argument
bc†
has type
TorchLean.NN.Seq
(Spec.Shape.scalar.appendDim 30)
(Spec.Shape.scalar.appendDim outDim)
but is expected to have type
?m.60 at bc†
(Spec.Shape.scalar.appendDim 29) τ
in the application
TorchLean.NN.compAny at bc†

despite annotations this will always compile, run, and fail

```
class Mlp(nn.Module):  
    def __init__(self):  
        super().__init__()  
        self.layer1: nn.Module = nn.Linear(10, 30)  
        self.layer2: nn.Module = nn.Linear(29, 1)  
  
    def forward(self, x):  
        x = F.relu(self.layer1(x))  
        x = self.layer2(x)  
        return x
```




“but annotations are annoying and useless i dont want to write them and duck typing is so convenient because flexibility or whatever”

Big duck propaganda!

- Useless annotations have been solved since the 70s, but Java wasn't paying attention
(see ML, Rust, TypeScript, Kotlin...)
- Structural type systems allow for “if it quacks like a duck...” typing
(again, see TypeScript)

Back to reality

- I'm not seriously advocating for using Lean to code neural networks
- However, stronger and flexible type systems are possible
- A man can hope for a TypeScript-like revolution for Python (and PyTorch!)



**static typing: awesome
(and TorchLean is pretty
cool)**



dynamic typing: stinks

Want to find the rest of the code?

<https://github.com/Tarekun/TorchLeanPresentation>

The screenshot shows the GitHub repository page for **TorchLeanPresentation** by user **Tarekun**. The repository is public and has 1 branch and 0 tags. The commit history shows 9 commits. The file list includes:

File	Description	Time
lean-mlp	reviewed dataset generation function	10 minutes ago
presentation	included presentation	4 minutes ago
pytorch	realigned hyperparameters	5 minutes ago
.gitignore	included presentation	4 minutes ago
README.md	adds links to readme	2 minutes ago

The **README** file is selected and shows the following content:

This is a quick demo project I wrote for a presentation given on fully typed checked neural networks using [TorchLean](#).

The project contains the implementation of a small MLP with one single hidden layer of 30 neurons trying to predict a $f: \mathbb{R}^{10} \rightarrow \mathbb{R}$ defined as $x \mapsto (\sum_i x_i)^2$. Data for training is generated by sampling a random batch points from $\mathbb{R}^{10}$ and Gaussian noise is applied to the value.

The `pytorch` directory contains an implementation of this neural network using the de-facto standard library PyTorch.

More interestingly, the `lean-mlp` directory contains the implementation of the same network and experiment using [TorchLean](#), a library that adds support for torch in the theorem proving language [Lean](#).

rant over